DEPARTMENT OF MATHEMATICS

UNIVERSITY COLLEGE OF ENGINEERING (A)

OSMANIA UNIVERSITY, HYDERABAD, TELANGANA–07

B.E (CSE) – IV–SEMESTER– CLASS TEST –II

ENGINEERING MATHEMATICS–IV

(PROBABILITY &STATISTICS)

Academic Year: 2025-2026
Date: 05-05-2026

Max Marks: 30
Duration: 1hr 30 min

	PART A (5x2=10 M) Answer All Questions	M	BT	CO
1. a	State the multinomial theorem.	2	2	3
b	Calculate mean and variance of normal variate.	2	3	3
c	Define null hypothesis and alternate hypothesis.	2	1	5
d	Write applications of Chi-square distribution.	2	2	5
f	Explain the critical region, degrees of freedom and level of significance.	2	2	5
	PART B (2x10= 20 M) Answer any TWO Questions			
2. a	A university conducts placement examination to all incoming freshmen. The examination scores of the 1000 examinees last semester were approximately normally distributed with mean of 80 and standard deviation of 5. Find the total number of examinees that randomly chosen student got a scores below 70? Above 82? Between 75 and 90? $P(0 \le z \le 0.4) = 0.1554, P(0 \le z \le 2) = 0.4772, P(0 \le z \le 1) = 0.3413$	5	4	3
b	Explain Exponential distribution and calculate its mean and varience.	5	4	3
3. a	In a large lot of electrical bulbs, the mean life and standard deviation of the bulbs are 360 hrs and 90 hrs respectively. A sample of 625 bulbs choosen. It is found that the mean life and standard deviation of the bulbs are 355 hrs and 90 hrs respectively. Can we conclud that the sample is drawn from the given population? Test at 5% level of significance and find the 95% confidence limits of the mean.(t=1.96)	5	4	5
b	Fit a Poisson distribution for the following data and test the goodness of fit at 5% level of significance. $(\chi^2(0.05) = 7.815)$	5	3	5

x	0	1	2	3	4	5
f(x)	110	170	130	60	23	7

4. a	Define Gamma distribution and calculate its mean and varience.											5 3 3
b	The values in two random samples are given below.											5 4 5
Sample1	15	25	16	20	22	24	21	17	19	23		
Sample2	35	31	25	38	26	29	32	34	33	27	29	31
	Can we conclude that the two samples are drawn from the same population? Test at 5% level of significance.(F=3.11)											